

The identity type $\text{Id } A a b$ with **Ref** and **J**

typing, reduction, injectivity, and subject reduction

Agda development ID/ (domain-semantics)

This note records the rules for the element-level Martin-Löf identity type added to the type-in-type theory ($\mathbf{U} : \mathbf{U}$) of the ID/ development, and states the two metatheorems proved for it. Everything below is machine-checked in Agda (`--without-K --exact-split`) with no postulates, no holes, and no termination/positivity pragmas.

The eliminator is Martin-Löf's *original* **J** with a fully dependent motive. Writing the motive and base types out,

$$C : (x y : A) \rightarrow \text{Id } A x y \rightarrow \mathbf{U}, \quad d : (x : A) \rightarrow C x x (\text{Ref } x),$$

so that $\text{J } C d p$ at $p : \text{Id } A a b$ has type $C a b p$ (the application spine $((C a) b) p$).

Typing rules

$$\begin{array}{c}
 \text{ID} \\
 \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma \vdash a : A \quad \Gamma \vdash b : A}{\Gamma \vdash \text{Id } A a b : \mathbf{U}}
 \end{array}
 \qquad
 \begin{array}{c}
 \text{REF} \\
 \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma \vdash a : A}{\Gamma \vdash \text{Ref } a : \text{Id } A a a}
 \end{array}$$

$$\begin{array}{c}
 \text{J} \\
 \frac{\Gamma \vdash C : (x y : A) \rightarrow \text{Id } A x y \rightarrow \mathbf{U} \quad \Gamma \vdash d : (x : A) \rightarrow C x x (\text{Ref } x) \quad \Gamma \vdash p : \text{Id } A a b}{\Gamma \vdash \text{J } C d p : C a b p}
 \end{array}$$

Conversion rules

$$\begin{array}{c}
 \text{J-}\beta \\
 \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma \vdash a_0 : A \quad \Gamma \vdash C : \dots \quad \Gamma \vdash d : \dots}{\Gamma \vdash \text{J } C d (\text{Ref } a_0) = d a_0 : C a_0 a_0 (\text{Ref } a_0)}
 \end{array}$$

J fires only on a *literal* $\text{Ref } a_0$, whose type is the diagonal $\text{Id } A a_0 a_0$; both sides of **J- β** therefore have the *same* type $C a_0 a_0 (\text{Ref } a_0)$, so the rule carries no endpoint or motive equalities and needs no injectivity. The congruences are

$$\begin{array}{c}
 \text{ID-CONG} \\
 \frac{\Gamma \vdash A = A' : \mathbf{U} \quad \Gamma \vdash a = a' : A \quad \Gamma \vdash b = b' : A}{\Gamma \vdash \text{Id } A a b = \text{Id } A' a' b' : \mathbf{U}}
 \end{array}
 \qquad
 \begin{array}{c}
 \text{REF-CONG} \\
 \frac{\Gamma \vdash a = a' : A}{\Gamma \vdash \text{Ref } a = \text{Ref } a' : \text{Id } A a a}
 \end{array}$$

$$\begin{array}{c}
 \text{J-CONG} \\
 \frac{\Gamma \vdash C = C' : \dots \quad \Gamma \vdash d = d' : \dots \quad \Gamma \vdash p = p' : \text{Id } A a b}{\Gamma \vdash \text{J } C d p = \text{J } C' d' p' : C a b p}
 \end{array}$$

(Grey premises re-typing the components are elided; see ID/Syntax/Typing.agda.)

Head reduction

Reduction is untyped weak head reduction. The single-step relation $M \rightsquigarrow_1 N$ has the following **ld**-clauses (alongside the usual β and application-congruence for functions):

$$\begin{array}{c} \text{J-RED} \\ \hline \text{J } C d (\text{Ref } a) \rightsquigarrow_1 d a \end{array} \qquad \begin{array}{c} \text{J-SCRUT} \\ \hline p \rightsquigarrow_1 p' \\ \hline \text{J } C d p \rightsquigarrow_1 \text{J } C d p' \end{array}$$

J-RED is the ι -step; J-SCRUT lets reduction proceed into the proof position so a non-canonical proof can reduce toward a Ref. Each step is a typed conversion (J-RED $\mapsto$ J- β , J-SCRUT $\mapsto$ J-cong), and $\rightsquigarrow_1$ is deterministic.

Domain semantics of **ld** and **J**

The model is the poset of *finite elements* $u, v, w, \dots$ of a Scott domain, with an information order $u \leq v$, a bottom $\perp$, a universe code $\mathbb{U}$, function elements, and Π -codes. The identity type adds two shapes:

$\mathbf{Id}(t, u, v)$ (the *code* of a type **ld** $A a b$: t codes A , u the value of a , v the value of b),

$\mathbf{r}(w)$ (a *proof value*: a witness w , the model's analogue of λ for functions).

Semantic membership is written $u \in a$ (“the finite element u is a member of the type-code a ”). The characteristic clause for the identity type is

$$\mathbf{r}(w) \in \mathbf{Id}(t, u, v) \iff w \in t \wedge w \leq u \wedge w \leq v,$$

i.e. a proof of $\mathbf{Id}(t, u, v)$ is exactly a witness of the underlying type that lies *below both endpoints* — the domain-theoretic reading of “ u and v agree down to w ”. All mismatched shapes (e.g. $\mathbf{r}(w) \in \Pi(\dots)$ or $\mathbf{Id}(\dots) \in \Pi(\dots)$) are empty, so identity codes and proof values interact only with one another.

Evaluation relates a term M , an environment ρ , and a finite element below its value; write $u \triangleleft \llbracket M \rrbracket_\rho$ for “ u approximates $\llbracket M \rrbracket_\rho$ ”. The new clauses (for $u \neq \perp$, where $\perp \triangleleft \llbracket M \rrbracket_\rho$ always holds) are

$$\begin{aligned} \mathbf{Id}(t, u, v) \triangleleft \llbracket \text{Id } A a b \rrbracket_\rho &\iff t \triangleleft \llbracket A \rrbracket_\rho \wedge u \triangleleft \llbracket a \rrbracket_\rho \wedge v \triangleleft \llbracket b \rrbracket_\rho, \\ \mathbf{r}(w) \triangleleft \llbracket \text{Ref } a \rrbracket_\rho &\iff w \triangleleft \llbracket a \rrbracket_\rho, \\ c \triangleleft \llbracket \text{J } C d p \rrbracket_\rho &\iff \exists w. w \triangleleft \llbracket p \rrbracket_\rho \wedge \text{br}(c, w), \quad \text{with} \quad \text{br}(c, \mathbf{r}(w')) \iff \text{fun}(w' \mapsto c) \triangleleft \llbracket d \rrbracket_\rho. \end{aligned}$$

So **ld** evaluates componentwise to an **Id**-code, **Ref** to a proof value, and **J** $C d p$ dispatches on the proof: once p evaluates to a proof $\mathbf{r}(w')$, the value of **J** $C d p$ is $\llbracket d \rrbracket$ applied to the witness w' (the single-edge condition “ d maps w' to c ”). **The motive C does not occur in the value** — it constrains only the *type*. This is the based-J computation rule read in the model.

The logical relation and adequacy

Adequacy is proved through a Kripke logical relation between the syntax and the model. For each type it defines, at every finite element, when a term is *valid*; write $\Gamma \Vdash M : A [u \in a]$ for “ M is valid at type A and code a , with value u ”. At an identity type the relation unfolds to:

$$\Gamma \Vdash M : \text{Id } A a b [\mathbf{r}(w) \in \mathbf{Id}(t, u, v)] \iff M \rightsquigarrow^* \text{Ref } w_0 \text{ for some } w_0 \text{ with } \Gamma \vdash w_0 = a : A \text{ and } \Gamma \vdash w_0 = b : A \text{ (both)}$$

That is: a valid proof *head-reduces to a canonical* $\text{Ref } w_0$ whose witness w_0 is validly equal to both endpoints. (Because proofs are proof-irrelevant, the underlying **Id**-code carries no selection edge; the two endpoint equalities are the whole content.)

Adequacy. By induction on derivations:

$$\Gamma \vdash M : A \implies M \text{ valid}, \quad \Gamma \vdash M = N : A \implies M, N \text{ validly equal.}$$

The J case (the based-J driver). Suppose $\Gamma \vdash \text{JCD} p : C a b p$ and let c be a finite element below $\llbracket \text{JCD} p \rrbracket_\rho$.

1. By the evaluation clause, c comes with a proof value: p evaluates to some $\mathbf{r}(w')$ and $\mathbf{r}(w') \in \mathbf{Id}(t, u, v)$, so $w' \leq u$ and $w' \leq v$.
2. The induction hypothesis for p at $\text{Id } A a b$ gives, via the relation above, a head reduction $p \rightsquigarrow^* \text{Ref } w_0$ with w_0 validly equal to a and to b . Hence

$$\text{JCD} p \rightsquigarrow^* \text{JCD}(\text{Ref } w_0) \rightsquigarrow_1 d w_0 \quad (\text{J-scrut}^* \text{ then J-}\beta).$$

3. *d-edge*. The induction hypothesis for d (a valid function), applied to the witness w_0 , makes $d w_0$ valid at the diagonal type $C w_0 w_0 (\text{Ref } w_0)$.
4. *Type transport*. Using $\Gamma \vdash w_0 = a : A$, $\Gamma \vdash w_0 = b : A$ and $\text{Ref } w_0 = p$, the syntactic types are convertible, $\Gamma \vdash C w_0 w_0 (\text{Ref } w_0) = C a b p : \mathbf{U}$; combined with monotonicity of the type-value along the information order, this carries the validity of $d w_0$ to the goal type $C a b p$.
5. *Head expansion*. Finally, validity is stable under the head reduction $\text{JCD} p \rightsquigarrow^* d w_0$ from step 2 (the contractum is valid, and the redex converts to it by J- β), so $\text{JCD} p$ is valid at $C a b p$. $\square$

The two-substitution (equality) version of this argument handles J-cong; it splits the two witnesses coming from the two sides and bridges them with the endpoint equalities.

Metatheorems (corollaries of adequacy)

Both results are obtained by running adequacy at the *bottom environment* and reading the logical relation off the model.

Theorem 1 (Id-injectivity, ID/IdInjectivity.agda). *If $\Gamma \vdash \text{Id } A_0 a_0 b_0 = \text{Id } A_1 a_1 b_1 : \mathbf{U}$ then*

$$\Gamma \vdash A_0 = A_1 : \mathbf{U}, \quad \Gamma \vdash a_0 = a_1 : A_0, \quad \Gamma \vdash b_0 = b_1 : A_0.$$

Evaluate the given conversion at the bottom environment: it yields a validated equality at the type-code $\mathbf{Id}(\perp, \perp, \perp)$, whose stored data are exactly the component conversions between domain, left and right endpoints; a head-reduction uniqueness lemma pins the codes, giving the three conversions.

Theorem 2 (Subject reduction, ID/SubjectReduction.agda). *If $\Gamma \vdash M : A$ and $M \rightsquigarrow_1 N$ then $\Gamma \vdash N : A$.*

For the J-RED step the contractum da_0 already has type $Ca_0a_0(\text{Ref } a_0)$, retyped to the goal $Cab(\text{Ref } a_0)$ using $a_0 = a$ and $a_0 = b$ extracted from $\text{Ref } a_0 : \text{Id } A a b$. For J-SCRUT the result type $Cabp$ is transported along $p = p'$, obtained from reduction-is-conversion; the Id -typed proof needs only the diagonal J- β , never Id -injectivity. The β -step for functions inverts λ -typing through Π -injectivity, as in the base theory.

The adequacy theorems are in `ID/Adequacy/Value.agda` (`adequacySub2 / adequacyEqSub2`); the J driver of steps 1–5 above is `ID/Adequacy/JApp.agda` (`value`), `JAppCross.agda` and `JAppCongr.agda` (`equality`).